



A COMMUNITY OF CHOICE

CITY ENGINEER'S OFFICE
2101 O'Neil Avenue
Cheyenne, Wyoming 82001

MEMO

TO: Honorable Mayor Patrick J. Collins, City of Cheyenne
Members of the City Council, City of Cheyenne

FROM: Wesley D. Bay, P.E., Deputy City Engineer

DATE: October 10, 2022

SUBJECT: Professional Service Agreement between the City of Cheyenne and Roadway Asset Services, LLC., for the Pavement Condition Survey and Rehabilitation Strategy Project.

Please find the attached Professional Service Agreement between the City and Roadway Asset Services, LLC., for engineering services related providing pavement condition indices and analyses to develop pavement rehabilitation and maintenance strategies for all roadways in Cheyenne, which includes main roadways through City parks.

Engineering Staff evaluates fee proposals for engineering services based on time and material estimates for the specific project. Those costs are readily tracked and verified during design operations and the City only pays for the actual time expended. We believe this agreement will provide the required services to complete the design portion of the project.

The fees include services for project management, intersection re-design, topographical surveying, geotechnical investigations, utility investigation, traffic signal design, and plan creation. Funding for this project will come from the 2018 1% Specific Purpose Tax Fund (5th Penny). The total professional services will be a cost not-to-exceed \$149,503.00. We respectfully request your approval of this professional services agreement.

City Contract No. _____

**PROFESSIONAL SERVICES AGREEMENT
for
Pavement Condition Survey and Rehabilitation Strategy**

THIS AGREEMENT is made and entered into this _____ day of _____, 2022, by and between the City of Cheyenne, located at 2101 O'Neil Avenue, Cheyenne, Wyoming 82001 ("City"), and Roadway Asset Services LLC., 6001 West Palmer Lane, Suite 370-1102, Austin, TX 78727 ("Consultant").

RECITALS

In consideration of the mutual covenants and obligations herein expressed, it is agreed by and between the parties as follows:

1. **SCOPE OF WORK**

The Consultant agrees to provide services as described for the above-mentioned project, in accordance with the Scope of Work attached hereto as Exhibit A, incorporated by reference, and expressly made a part hereof.

2. **TIME OF PERFORMANCE**

This Agreement shall commence on the date the last required signature is attached hereto and shall continue in full force and effect until the scope of work is completed in accordance with Exhibit A, unless this Agreement is earlier terminated pursuant to its provisions.

3. **COMPENSATION**

In consideration of the services to be performed pursuant to this Agreement, the Consultant will bill the City and the City agrees to pay Consultant a not to exceed amount of One-Hundred Forty-Nine Thousand Five-Hundred and Three Dollars and Zero Cents (\$149,503.00).

4. **CONSULTANT RESPONSIBILITY**

In providing services under this Agreement, the Consultant shall perform in a manner consistent with that degree of care and skill ordinarily exercised by members of the same profession currently practicing under similar circumstances. Upon notice to the Consultant, the Consultant will, without additional compensation, correct those services not meeting such a standard.

5. **COMPLIANCE WITH LAWS**

The Parties agree that they will perform their obligations as provided in this Agreement in accordance with all applicable laws and ordinances.

6. **DRUG-FREE WORKPLACE**

In compliance with the Drug-Free Workplace Act of November 1988, the City of Cheyenne has established an Alcohol and Controlled Substance Policy that pertains to alcohol and drug usage by City Employees. All architects, engineers, and other consultants under

contract with City of Cheyenne, and their employees and sub-consultants, are required to comply with the provisions of the City's Alcohol and Controlled Substance Policy for drug and/or alcohol usage on City property or other sites occupied by the Consultant while performing the duties and responsibilities of the Agreement. It is the responsibility of the Consultant to familiarize themselves with the requirements of this policy and to inform all their employees and sub-consultants of the requirements and ensure their compliance. If the Consultant, their employees, or sub-consultants are found in violation of this policy, the Agreement may be terminated.

7. INDEMNIFICATION/HOLD HARMLESS

The Consultant agrees to indemnify, hold harmless and defend the City from and against any and all liabilities, claims, penalties, forfeitures and suits, and the cost and expenses incident thereto, including reasonable attorney's fees, which may hereafter arise as a result the performance of the Consultant's duties, including death or bodily injury to any person, destruction or damage to any property, contamination of or adverse effects on the environment, or any violation of governmental laws, regulations or orders to the extent caused by (1) the Consultant's breach of any term or provision of this Agreement; or (2) any negligent or wrongful act, error or omission by the Consultant, or its employees or sub-consultants in the performance of this Agreement. The Consultant acknowledges that it may incur a financial obligation to the City pursuant to the terms of this paragraph.

8. INSURANCE REQUIREMENTS

The Consultant must provide proof of the following insurance coverages:

Commercial General Liability Insurance

For claims arising out of bodily injury, illness, or death, or from damage to or destruction of property of others, including loss or use thereof, with minimum limits of \$1,000,000 per occurrence, \$2,000,000 aggregate for the entire term of the Agreement.

Workers' Compensation

Workers' compensation coverage shall be in effect for the entire term of the Agreement, as required by Wyoming law, for all employees or agents providing services under this Agreement. Consultant shall provide the City with proof of workers' compensation or employer's liability insurance coverage.

Professional Liability Insurance

The Consultant shall provide proof of professional liability insurance or errors and omissions liability insurance in an amount not less than \$500,000 to protect the City from any and all claims arising from the Consultant's negligence in the performance of duties under this Agreement. The City prefers that this liability insurance coverage be provided pursuant to an "occurrence" policy.

If this coverage is provided pursuant to a "claims made" policy:

- a. Consultant shall, concurrently with the execution of this Agreement, provide the City with a certificate of insurance demonstrating that such coverage is or shall be in effect at the time the Consultant begins the provision of services under this Agreement; and
- b. In the event the Consultant's services extend into a future policy period, the Consultant shall, prior to the policy expiration date, provide the City with a new certificate of insurance demonstrating that such coverage is or shall be in effect during all periods of time that Consultant will provide services under this Agreement; and
- c. Consultant shall maintain said "claims made" coverage for a period of five (5) years following the last date that Consultant has provided services under this Agreement; and
- d. In the event the Consultant or the insurer terminates "claims made" coverage prior to the expiration of the periods provided in subparagraphs a., b., or c. of this paragraph, the Consultant shall provide to the City advance written notification of the termination of said coverage and shall provide the City with an endorsement for an extended reporting period ("tail coverage") which shall be in effect for a period of time not less than five (5) years following the last date that Consultant has provided services under this Agreement.

Additional Insurance Information

The Consultant shall name the City of Cheyenne as **an Additional Insured** by endorsement on its insurance policies and shall provide the City with a copy of the endorsements. This requirement does not apply to workers' compensation and professional liability insurance policies.

Consultant shall provide the City with certificates of insurance acknowledging the above-stated coverages prior to beginning any work under this Agreement.

It is understood and agreed that these policies are primary and not contributory. All policies required under this Agreement shall be in effect for the duration of the Agreement. It shall be an affirmative obligation upon Consultant to immediately notify in writing the city risk manager, city clerk, and city attorney of any fact, circumstance, or occurrence that has resulted in or may result in the cancellation of, or substantive change to any insurance coverage required by this Agreement, and failure to do so shall be construed to be a breach of this Agreement.

If requested by the City, the Consultant shall provide the City with copies of insurance policies and/or policy endorsements listing the City of Cheyenne as an additional insured. The City's failure to request or review such policies, endorsements, or certificates shall not affect the City's rights or Consultant's obligation hereunder.

Any insurance company providing coverage under this Agreement shall have a minimum A. M. Best rating of A- (excellent).

9. MINORITY AND DISADVANTAGED BUSINESS ENTERPRISES 49 C.F.R. Part 26

All parties to this Agreement assure that no person will be excluded from participation in, denied the benefits of, or otherwise discriminated against, in connection with the award and performance of this Agreement on the grounds of age, race, color, disability, national origin or sex.

10. MONTHLY REPORT

With every monthly billing, Consultant shall provide the City Representative with a written statement of the status of the work with respect to the Scope of Work, time sheets, and work schedule. Failure to provide the required monthly report will delay processing of any payment request until the report is submitted.

11. INDEPENDENT CONSULTANT

The Consultant shall function as an independent consultant for the purposes of this Agreement. The Consultant shall assume sole responsibility for any debts or liabilities that may be incurred by the Consultant in fulfilling the terms of this Agreement. Nothing in this Agreement shall be interpreted as authorizing the Consultant or its agents or employees to act as an agent or representative of or on behalf of the City or to incur any obligation of any kind on the behalf of the City.

12. TAXES

The Consultant agrees to pay all valid taxes, excises, license fees, permit fees, bills, debts, and obligations incurred by and in connection with its operations under this Agreement.

13. DEFAULT

Each and every term and condition in this Agreement shall be deemed to be a material element of the Agreement. In the event either party should fail or refuse to perform according to the terms of this Agreement, such party may be declared in default.

14. REMEDIES

In the event a party has been declared in default, such defaulting party shall be allowed a period of fifteen (15) days within which to cure the default. In the event the default remains uncorrected, the non-defaulting party declaring default may elect to:

- a. Terminate the Agreement and seek damages, which damages shall not exceed the contract amount; or
- b. Treat the Agreement as continuing and require specific performance.

15. TERM AND TERMINATION

The term of this Agreement shall be from the date of execution and continue until the scope of work is completed in accordance with Exhibit A. The parties may by mutual written agreement renew or extend this Agreement. The City may, without cause, and upon thirty days' written notice to the Consultant, terminate this Agreement in whole or in part at any time, for the City's convenience. Upon receipt of such notice, the Consultant shall:

- a. Discontinue all services affected; and
- b. Deliver to the City within five (5) days all data, drawings, specifications, reports, estimates, summaries, and such other information and materials as may have been accumulated by the Consultant in performing this Agreement, whether completed or in process.
- c. In the event of termination for convenience, the City will pay the Consultant for accepted work done and expenses incurred to the date of termination. Such acceptance shall not be unreasonably withheld.

16. ADDITIONAL REMEDIES

In the event the Consultant fails to strictly perform in accordance with this Agreement, the City may elect to correct the deficiencies and charge the Consultant. In the event of default of any of the conditions by either party which shall require the party not in default to commence legal or equitable action against the defaulting party each party shall bear its own costs and expenses, including without limitation, attorney's fees.

17. GOVERNING LAW, JURISDICTION AND VENUE

The construction, interpretation and enforcement of this Agreement shall be governed by the laws of the State of Wyoming. The courts of the State of Wyoming shall have jurisdiction over any action arising out of this Agreement and over the parties, and the venue shall be the First Judicial District, Laramie County, Wyoming.

18. GOVERNMENTAL IMMUNITY

The City of Cheyenne, along with its officials and employees, does not waive its governmental immunity by entering into this Agreement, except to the extent necessary for the parties to pursue a contract action to clarify or enforce the written terms of the Agreement. Furthermore, the City specifically retains all immunities and defenses available to it as a sovereign or governmental entity pursuant to Wyo. Stat. § 1-39-101, *et seq.*, and all other relevant state and federal law. Designations of venue, choice of law, enforcement actions, and similar provisions should not be constructed as a waiver of governmental immunity. The parties agree that any ambiguity in this Agreement shall not be strictly construed, either against or for either party, except that any ambiguity as to governmental or sovereign immunity shall be construed in favor of governmental immunity.

19. MONITORING ACTIVITIES

The City shall have the right to monitor all activities related to this Agreement that are performed by the Consultant or its sub-consultants. This shall include, but not be limited to, the right to make site inspections at any time and with reasonable notice; to bring experts and consultants on site to examine or evaluate completed work or work in progress; to examine the books, ledgers, documents, papers, and records pertinent to this Agreement; and to observe personnel in every phase of performance of the related work.

20. OWNERSHIP OF DOCUMENTS AND WORK PRODUCT

All deliverables by the Consultant, all work product of the Consultant, all notes, calculations, memoranda, or any other documentation generated by the Consultant pursuant to the scope of this Agreement, shall be the property of the City of Cheyenne and shall be provided to the City of Cheyenne immediately upon request.

21. NON-DISCRIMINATION

The Consultant shall comply with the Civil Rights Act of 1964, the Wyoming Fair Employment Practices Act (Wyo. Stat. § 27-9-105, *et seq.*), the Americans With Disabilities Act (ADA), as amended, 42 U.S.C. § 12101, *et seq.*, and the Age Discrimination Act of 1975 and any properly promulgated rules and regulations thereto, and shall not discriminate against any individual on the grounds of age, sex, creed, color, race, religion, national origin, ancestry, pregnancy or qualifying disability in connection with the performance under this Agreement.

22. FORCE MAJEURE.

The performance of the Agreement by either party shall be subject to force majeure including, but not limited to, acts of God, fire, flood, natural disaster, war or threat of war, acts or threats of terrorism, civil disorder, unauthorized strikes, governmental regulation or advisory, recognized health threats as determined by the World Health Organization, the Centers for Disease Control, or local government authority or health agencies (including, but not limited to, the health threats of COVID-19, H1N1, or similar infectious diseases), curtailment of transportation facilities, or other similar occurrence beyond the control of the parties, where any of those factors, circumstances, situations, or conditions or similar ones prevent, dissuade, or unreasonably delay the performance required by this Agreement. The Agreement may be cancelled by either party, without liability, damages, fees, or penalty, and any unused deposits or amounts paid shall be refunded, for any one or more of the above reasons, by written notice to the other party.

23. PROFESSIONAL REGISTRATION

The Consultant shall endorse, as required by law, plans and reports prepared under this Agreement, and shall affix thereto his or her seal of professional registration, showing that he or she is licensed to practice in the State of Wyoming, if necessary.

24. ENTIRETY OF AGREEMENT

This Agreement consisting of seven (7) pages and Exhibit A consisting of twelve (12) pages contain the entire understanding of the parties. There are no other terms or conditions, written or oral, concerning or controlling this matter.

IN WITNESS WHEREOF, the parties have hereunto set their hands and seals the day and year first above written.

City of Cheyenne

(SEAL)

By: _____
Patrick J. Collins, Mayor

Attest:

Kristina F. Jones, City Clerk

Date: _____

Consultant

By: _____

Print Name: _____

City of Cheyenne Pavement Condition Survey and Rehabilitation Strategy



Prepared by:

Roadway Asset Services, LLC.

6001 W Parmer Lane

Suite 370-1102

Austin, Texas 78727

bwilliamson@roadwayassetservices.com

210-837-5249

Firm Overview

RAS Roadway Asset Services, LLC (RAS) is a Texas Engineering Firm (License #: F-22104) headquartered in Austin, Texas. RAS team members have managed automated data collection, performed QA/QC, developed pavement management plans, supplied GIS based deliverables, and provided imports and modeling for Cities across the United States. RAS offers comprehensive experience and subject matter expertise in the fields of engineering, surveying, asset management, transportation planning, and GIS. **The RAS executive team has managed over 100,000 miles of pavement condition surveys and ROW asset data in the last 5 years.** Additionally, the RAS team has extensive experience rating pavement per the ASTM D6433 methodology both via the sampling methodology and by utilizing artificial intelligence assisted by rigorous Quality Assurance protocols. The RAS leadership team consists of seasoned **pavement management professionals who bring over 100 years** of database integration and pavement condition survey management. **Our team will provide prompt and relevant services as demonstrated by RAS leadership's (Scot Gordon, Bart Williamson, and Rafael Rivera) involvement in the last data collection cycle.** Throughout their careers, the RAS leadership team has been involved in pavement condition and ROW asset surveys for similar projects including:



- Cheyenne, WY
- Salt Lake City, UT
- Charlotte, NC
- Sioux Falls, SD
- Albuquerque, NM
- Memphis, TN
- Denver, CO
- Mesa County, CO
- Colorado DOT
- Minnehaha County, SD
- Greenville, SC
- Durham, NC
- Grand Junction, CO
- San Antonio, TX
- Pennington County, SD



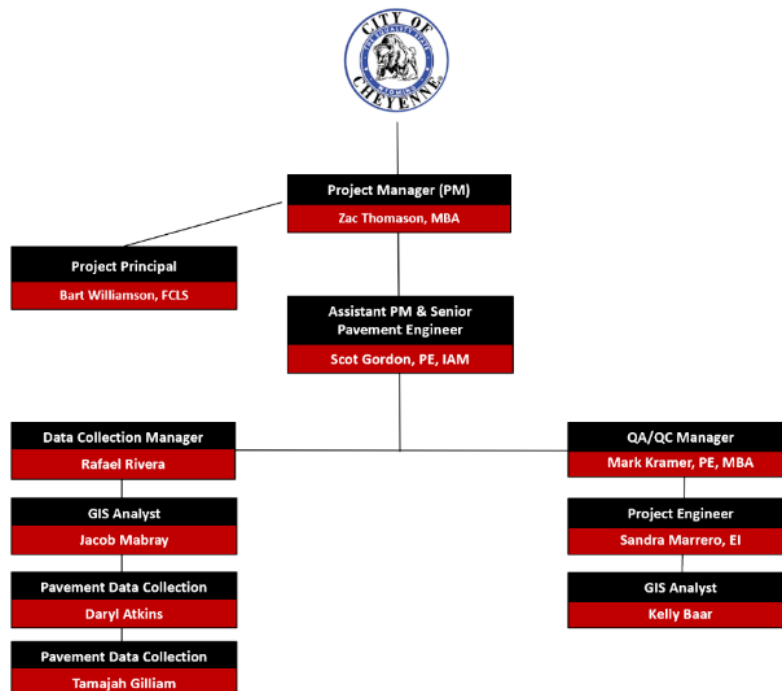
RAS has invested in the **most sophisticated** fleet of roadway asset collection (RAC) vehicles and pavement analysis tools for automated data collection that provides a **100% linear assessment** of the roads driven. **This methodology removes the subjectivity of rating small sample areas of the road segment.** Unlike many of our competitors, RAS utilizes a ROW capture system to provide an **immersive 360 view** versus stationary independent camera views. Furthermore, RAS' RAC vehicles have **received independent inertial profiler**

certification for accuracy and repeatability from Texas A&M Transportation Institute (TTI) and the National Center for Asphalt Technology at Auburn University (NCAT).

Project Team Overview

- **Project Manager, Zac Thomason, MBA** has nearly 17 years of experience in pavement management and has personally led over 350 condition studies. Mr. Thomason will deliver final reports and establish channels of communication (i.e. monthly/weekly meetings) between the RAS team and the City of Cheyenne.
- **Assistant Project Manager and Senior Pavement Engineer, Scot Gordon, PE, IAM** has over 30 years of experience in pavement engineering, asset management, and pavement design. Mr. Gordon will review distress interpretations and provide expertise in pavement management optimization. In addition, Scot will perform a 10-mile field pilot validation study to make necessary corrections to processing routines and review quality assurance measures with City staff.
- **Project Principal, Bart Williamson** has over 30 years of project management experience and will work with Mr. Gordon and Mr. Thomason to coordinate project needs, resources, and ensure high-quality deliverables are received in a timely manner.
- **Quality Assurance/Quality Control Manager, Mark Kramer, PE** has 24 years of experience in engineering and will review data to ensure 100% collection of the network. Mr. Kramer will coordinate with the RAS pavement evaluation team to ensure data collected from automated equipment and artificial intelligence correlates with visual ratings.
- **Data Collection Manager, Rafael Rivera**, has 12 years of experience and is responsible for updating schedules based on collection status, overseeing the daily operation crew, and initiating/ reviewing the evaluation process for pavement and ROW assets.
- **Project Engineer, Sandra Marrero, EI**, has 8 years of experience in evaluating pavement conditions, processing pavement ratings, preparing reports, and performing maintenance budget scenarios. Ms. Marrero will assist in the development of the final pavement report.

PROJECT ORGANIZATION AND STAFFING CHART



Proposed Methodology - Scope of Work

Kick-Off Meeting

At the outset of the project, RAS will work in conjunction with City Staff to review the City's existing GIS centerline files for use in this project. After an initial review, RAS will conduct a kickoff meeting with City staff to discuss the GIS files, deliverable formats, and additional data needed by RAS for preparation, process for interim review, and project administration. During the kickoff meeting, RAS will present a project schedule, quality assurance/quality control (QA/QC) plan, goals for the project, and the methodology for the pavement condition assessment (including expounding on the project approach, presenting equipment and its capabilities, and discussing staff roles/assignments). **In addition, team members will determine a pilot area of approximately 10 miles for collection and review of all deliverables by City Staff**, obtain a collection letter, and determine areas that are of greatest priority for collection or that require special timing for collection due to festivals such as **Annual Western Spirit Art Show & Sale, Fridays on the Plaza, and Annual End of the Trail Kite Festival**.



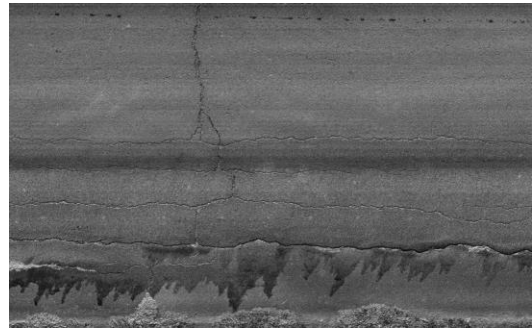
Pavement Condition Survey Equipment- Roadway Asset Collection (RAC) Vehicles

To complete the automated pavement condition survey, the RAS team will utilize RAC vehicles from the fleet of four (4), equipped with:

- The LCMS-2 camera is a downward-facing laser array providing images used to evaluate data that conforms with ASTM D6433 protocols, which uses **two 1-millimeter-pixel** resolution line scan cameras to **provide a customized digital condition rating system to collect user defined severity/extent-based pavement distresses and rutting**.
- The pavement distress type, density, severity, and extent are collected with the LCMS-2 and are used to calculate a Pavement Condition Index (PCI) score, between 0-100, that represents the condition of 100% of the driven lanes.
- Point Gray Ladybug 5+ 32MP **360-degree High-Definition camera** (utilized for accurate ROW asset capture, extraction, and pavement QA/QC) is far superior to multiple independently mounted HD cameras.
- **Linear distance measuring to within +/-0.5%.**
- A class 1 inertial profiler for simultaneously capturing dual-wheel path (left and right) International Roughness Index (IRI) measurements to the hundredth inch, in accordance with AASHTO R48. The profiler has gone through **ASTM E-950** certification and **has been independently certified by Texas A&M Transportation Institute (TTI) and the National Center for Asphalt Technology (NCAT)**. The inertial profiler meets the requirements and will be operates in accordance with **AASHTO Standards M 328, R 57-10, R 56-10, and R43M/R43-7**.
- Applanix POS/LV with DGPS (**Provides accurate internal GPS navigation for geo-locating pavement and right of way asset information**).



All subsystems for the RAC vans are integrated using tight synchronization between all data streams on the truck in real-time, referenced to both time and distance. All sensor locations are coordinated to the vehicle's reference point, together with the GPS and IMU hardware, using 3D translations and rotations. This allows the final world coordinates of all sensor data streams to be calculated and integrated. The methods for automated data collection and pavement condition rating are repeatable and defensible.



The RAS team will conduct pavement survey work on dry pavement and in lighting conditions that ensure accurate crack detection. We will collect imagery during daylight hours only, with no rain, fog, or snow visibility obstructions. Any road segment(s) that exhibits low image quality due to lighting will be recollected at a later time. The industry is currently migrating toward automated intelligence (AI) rating for 100% linear assessment of the roads, in accordance with guidelines within the **ASTM D6433, ASTM E1656, and ASTM E3303**. This methodology removes the subjectivity of rating small sample areas of the road segment where the sample may not represent the condition of the entire roadway segment.

Detailed Quality Assurance & Quality Control Work Plan

While the 10-mile pilot reviewed by RAS & City Staff will illustrate the completeness and accuracy of the RAS data stream, RAS has also developed a detailed 30-page **Data Quality Management Plan (DQMP)** to provide our clients with a systemized method for assuring data is representative of the conditions present. Included in the DQMP is a description of condition survey procedures, data collection vehicle and system calibration/verification, definitions for appropriate weather conditions, range of accuracy, data checks, safety procedures, verification by trained engineers, roadway segment review/verification, and integration into asset management programs. **We can**



supply the City of Cheyenne with our published 30-page DQMP at the request of City Staff. The document's length and technical nature didn't make it suitable for inclusion with this proposal submission.

Quality Assurance Step #1 – Network Database Review (Completeness / Gap Analysis)

While a simple routine in every project, the cornerstone to project initiation is conducting a complete diagnostic of the roadway network, including a full and thorough assessment of the City's GIS centerline. This will include a review of the City's data requirements (i.e., what information is needed/desired) and subsequent data gap analysis (what is missing).

RAS will include the following in this initial review:

- a. Base inventory information, (i.e., Functional Class, lengths, areas, surface type, etc.)
- b. Historical condition information
- c. Status of survey history
- d. Construction and maintenance history (optional task)
- e. Review of maintenance and rehabilitation treatments and unit costs (optional task)
- f. Review of existing pavement deterioration curves (optional task)

Proprietary Information

Quality Assurance Step #2 – Field Pilot

RAS will work with the City to determine a field pilot area of approximately 10 miles for collection and review of all deliverables by City Staff. The importance of the **field pilot** cannot be understated as it has become a routine milestone for the RAS Team on all pavement condition projects. The pilot allows RAS to collect, process, and review condition data with City Staff to ensure accuracy with the data collection and interpretation protocols. **The review of the RAS condition data will be hosted by Assistant Project Manager and Senior Pavement Engineer, Scot Gordon, PE, IAM in the field, where he will review site conditions with City Staff.** The importance of this step is to make necessary corrections to the processing routines that result from local standard construction practices/soil conditions and to ensure accuracy with the approach.



Quality Assurance Step #3 – Subsystem Monitoring In The Field



During the survey, the collection software monitors the GPS subsystems and alerts the operator if the GPS feed drops out or if GPS quality is compromised. The IMU will provide acceleration-based corrections during this time to ensure that GPS accuracy is maintained as much as possible.

The RAS team will also review a randomized sample of images to ensure that it complies with the requirements of the specifications throughout the course of the network matching and event QC. Upon completion of the network matching, an image report shall be generated with the total image count compared with that expected for each road.

Daily Progress Reports are produced by uploading sensor, GPS, and event data and matching against the road network definition. Progress reports include the following, road sections collected, length discrepancies, and remaining sections to be completed.

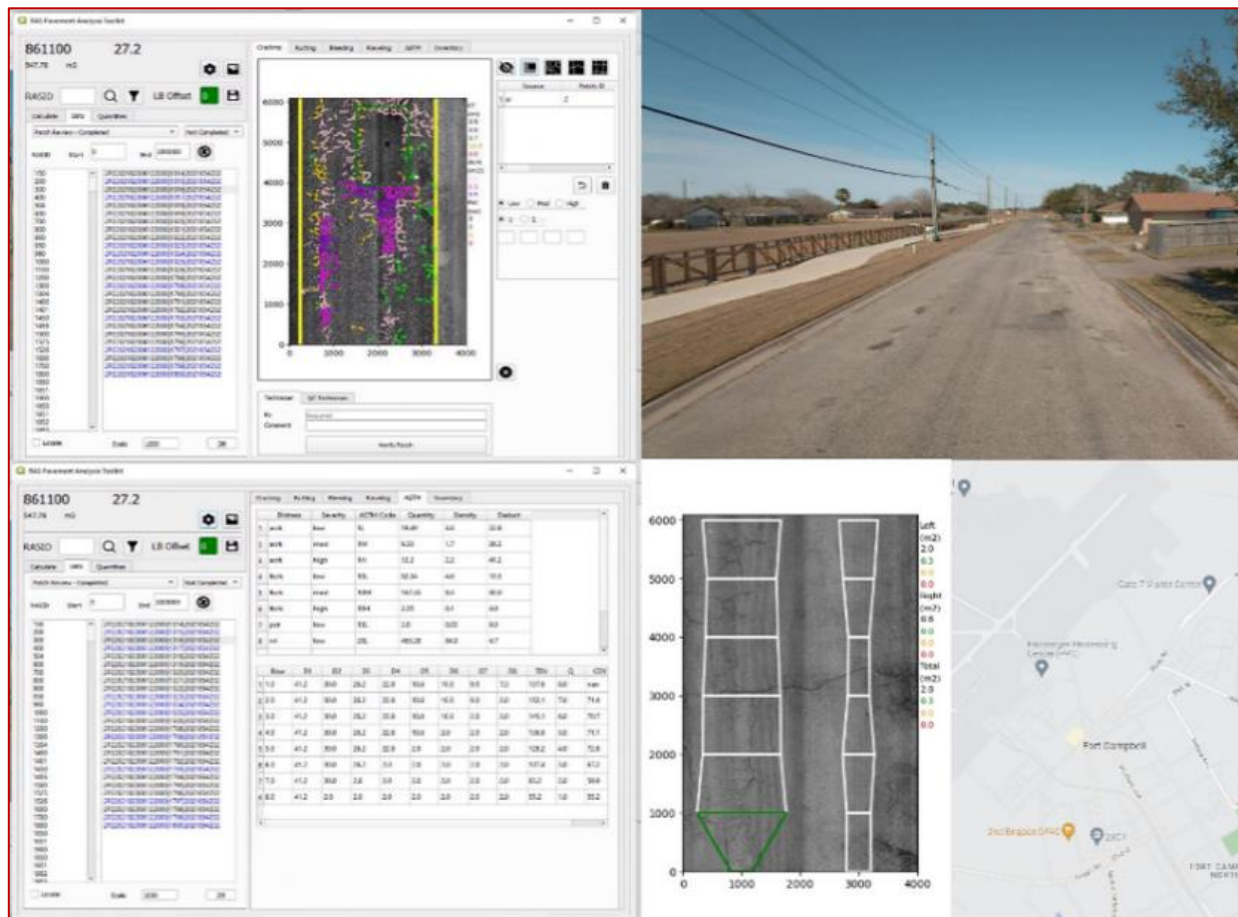
Quality Assurance Step #4 – Automated Crack Analysis & Engineer Review

After data is collected in the field and uploaded to the office environment, it is imported using the RAS AI pavement rating tool **Road TRIP™ (Technical Rating Intelligence Program)**. The import process creates mappings to the data so that users do not need to keep track of where the data is stored on central data server(s). At this stage, the major data processing tasks also occur, such as generation of right-of-way and pavement image streams; calculation of profile, roughness, rutting, detection of cracks, lane-markings, man-made objects, and other distresses. The automated crack analysis detects cracks which are overlaid on the pavement images and offset to assist with the verification of the detected cracks. The longitudinal and transverse crack distresses, that are transformed from the LCMS detected cracks, are used during reporting to identify where the LCMS detected cracks are found to result in an unacceptable level of false positives. These affected distress cracks are eventually deleted. During reporting, the



distress cracks are defined by road zone and accumulated according to the units defined in the client specification. The severity levels are identified based upon the defined limits (ASTM D6433) and verified for resolution through visual quality control checks of image files. Where density metrics are required, these are determined using the length of the interval being reported and the width of road zones included.

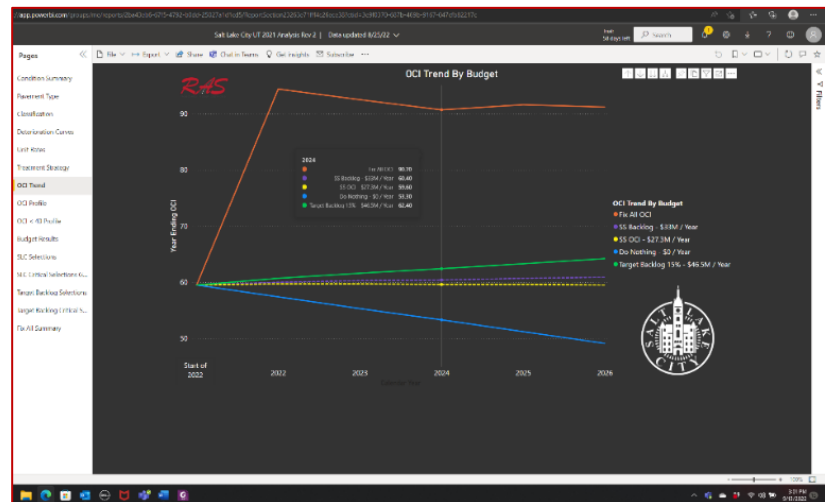
An experienced pavement inspector will perform QC to confirm the distresses and severity of the pavement condition data collected by the automated technology. This manual quality review is performed, in accordance with the principles of the ASTM D6433 standard, using the LCMS pavement images gathered during collection with the distresses superimposed and color coded, such as what can be seen in the image below.



Pavement Analysis, Budgetary Modeling, & Multi-Year Plan Optimization

RAS can assist the City with both inventory and condition assessment to enhance the level of service for its business units. RAS can also assist the City with determining the right treatment (prescription) at the right time. The RAS team has vast experience in capital planning (decision trees) of these inventoried assets as they are strategically programmed for replacement, refurbishment, or general maintenance.

Project Manager, Zac Thomason will utilize RAS' Budget Optimization Street Selector tool (**BOSS™**) to adopt the City's segment level GIS centerline, run budgetary models and forecast to establish trends, financially optimize the multi-year plan using sound "cost of deferral" constraints, and produce a final maintenance and rehabilitation plan for the City's review. **BOSS™** is a cloud based application with



powerful pavement management algorithms behind it that export the results of the pavement analysis to a user friendly interface such as **Microsoft PowerBI**, **ESRI storyboard maps**, or even a simple **Excel spreadsheet** if desired. All of the results are integrated with the City's existing GIS and supplied to the City as a **Personal/File Geodatabase** for consumption and storage.

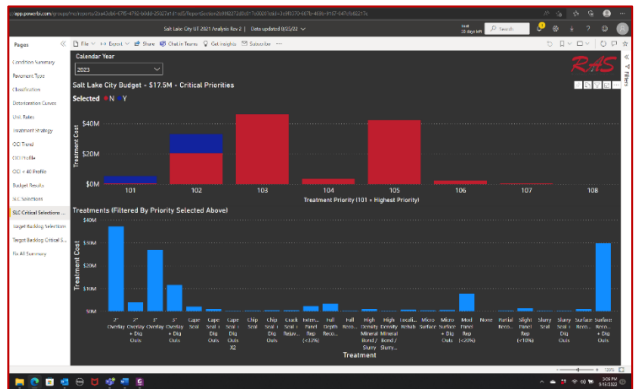
RAS will work with the City to establish the maintenance and rehabilitation activities, PCI trigger points, costs, reset PCI values, completed rehabilitation work since the survey, planned work, existing budgets, pavement deterioration curve development/assignment, and inflation priorities. Assistant Project Manager, Scot Gordon, will assist City Staff with determining the right treatment at the right time by reviewing the City's existing maintenance and rehabilitation strategies and recommending others that may be a good fit. The scope will include developing up to 8 multi-year pavement management scenarios using different philosophies, budget levels, and distributions. While RAS will define the scenarios to be run with City staff, at a minimum the following questions should be answered with the scenarios:

- **What is the resultant network PCI at my current funding level?**
- **What budget is required to maintain my existing network PCI?**
- **What budget is required to achieve a desired network PCI?**
- **What budget is required to control the growth in backlog?**

RAS will also consult with City staff to develop models utilizing different types of rehabilitation strategies (worst first, best first, most economic, need year, etc.). RAS understands that getting buy-in from City Administrators and Elected Officials means developing a long-range rehabilitation plan that considers local priorities. The RAS team will ensure that already approved rehabilitation work is programmed into the budgetary models for selection during the run. In addition, RAS will consult with City staff to identify the total cost (mill, overlay, traffic control, striping, etc.) of each rehabilitation method.

RAS' approach to such an analysis involves the following sequences and engagements during the process:

- Maintenance & Rehabilitation Setup** – to ensure the results of the budget model runs meet the City's expectations, RAS will discuss current Maintenance and Rehabilitation setup within the pavement analysis setup. This will include activity set up, PCI trigger points, and PCI reset scores.
- Deterioration Curves** – forecasting pavement conditions requires a detailed set of pavement deterioration curves for each roadway traffic classification and pavement material type, as designated by the pavement management system.
- Project development** – the BOSS™ analysis includes stitching segments (block) together to form a project, also known as a “management section”. RAS will work with City Staff to review the initial model results and begin “stitching” segments together to form logical projects that best meet the needs of the City.
- Financial Optimization & Prioritization** – RAS' BOSS™ analysis uses sound engineering and economic logic to prioritize which street candidates are selected throughout the multi-year plan. While most pavement management programs will prioritize by roadway traffic and condition, a **BOSS™** analysis takes it a step further and introduces **financial optimization into candidate selection** through the use of a “Need Year” analysis that identifies each segment's cost of deferral. Understanding the “Cost of Segment Deferral” allows the analysis to maximize the City of Cheyenne's limited funds in the best manner possible.

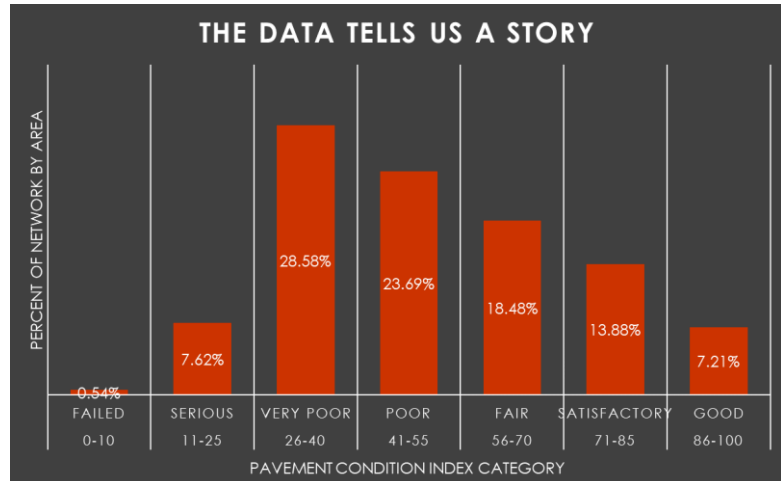


Pavement Final Report

RAS can provide the City a final report with statistical charts, graphs, and area maps illustrating all PCI results, the overall City's road quality, and findings from the pavement evaluation. This report will also include recommended updates to pavement performance models and deterioration calculations to reflect new and historical data. This will require feedback from the City and working sessions to develop the most accurate models for the administration of a maintenance plan.

We will use the collected data, along with available historical condition data, to develop and model deterioration curves that can be used to estimate the remaining pavement service life of all studied roadway sections. The curves will be presented in the report and provided in an Excel spreadsheet for future use by the City staff. The report will be provided for review by City staff and modified based on comments to produce a final report which will be delivered as a bound report and in Microsoft Word (.doc and .docx) and Adobe (.pdf) format.

Additionally, Scot looks forward to visiting with City staff on other **innovative considerations** to pavement management such as **including PCI of curb and gutter in selecting roads for maintenance; whether or not to deduct for potholes if the City has a pothole repair crew; how to get all council districts above a specific PCI threshold; and focused analysis on top traveled corridors where you are improving the level of service on the roads receiving the greatest traffic while also analyzing the impacts to the remainder of the network.**



For example, in the City of Fort Worth, Scot worked with City Staff to incorporate curb and gutter as a factor in street selection for maintenance. In addition, the previous evaluation deducted for potholes in their PCI score, while the most recent survey reversed this decision as their crews were repairing these distresses daily and a preference to not have potholes influence the PCI value were expressed.

In another example of innovation for the City of San Antonio, Scot performed additional analysis scenarios to identify variable funding needs to achieve and maintain the set goals for each Council District (minimum PCI of 70), based on pavement condition performance. To accomplish this, Mr. Gordon worked with the City to determine the proper mix between preservation and reconstruction within each of the 10 Council Districts to achieve the minimum baseline number of a PCI of 70 per district.

City Council Presentations

The RAS team can provide a public presentation to the City Council, including data, charts, and all summary data to showcase the results of the pavement condition survey. Scot Gordon has extensive experience preparing and presenting results of pavement and asset conditions assessments, as well as pavement budget scenarios and maintenance



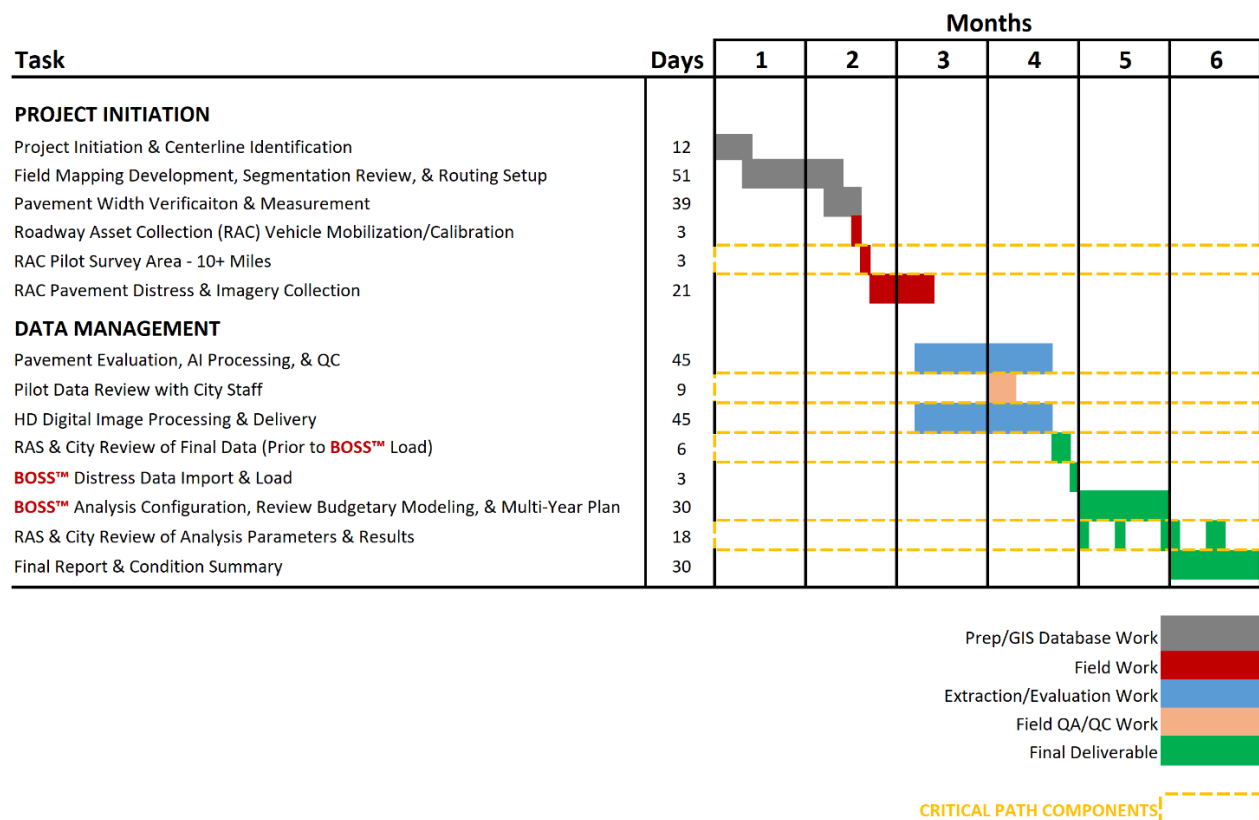
plans to Councils and Commissions as final reports. This is a standard scope that we provide to all our asset management clients. Scot has worked with County Commissions or City Councils with several agencies including Durham, NC; Pitkin County, CO; Pflugerville, TX; San Antonio, TX; Brownsville, TX; Indianapolis, IN; and Laredo, TX to name a few.

Project Schedule

RAS has assigned its Executive Leadership Team to this project to ensure the successful completion of the Roadway Network Condition update. RAS embraces a **Critical Path Method (CPM)** to project scheduling which includes building out a project roadmap to help visualize what needs to be done to reach the project end goal. The RAS project management technique involves mapping out key milestones or critical tasks necessary to complete a project including kickoff meeting, GIS data review, Data Collection, field pilot validation, pavement condition assessment, final deliverables and areas involving City input to avoid bottlenecks. RAS will work with the City in establishing key milestones as part of the CPM.

Due to the winter season approaching, data collection will be scheduled for late winter or early spring (late March or early April) to avoid the coldest of temperatures and optimize daily field production. The RAC Laser Crack Measurement System has an automated cold weather shut down procedure that generally initiates around 32 degrees Fahrenheit to protect its internal components.

The RAS team collects an excess of 30 pavement/ROW assessments annually and has an outstanding reputation of completing projects of similar size and scale on time and within budget. Progress meetings will be held at increments established with the City, to ensure clear expectations, facilitate open communication, and ensure mutual understanding of the project requirements and deliverables.



Project Fee Schedule

RAS team members have managed, provided field data collection, performed QA/QC, and developed pavement management plans on numerous similar projects to the scope of services requested by the City. RAS is a financially secure limited liability company that practices strong internal controls, conservative business practices, and a commitment to continued financial growth.

The RAS fee structure for this assignment can be found below with itemized tasks to illustrate the full scope of this project. The RAC vehicle will survey the arterial and collector roadways in two directions while surveying the residential roads in a single direction, resulting in approximately 500 survey miles. This mileage accounts for new roadways that currently exist in the City's GIS but do not yet exist in the City's current PAVER database.

City of Cheyenne, WY Pavement Condition Survey 				
Task	Description	Units	Unit Cost	Fee
	Required Services			
1	Project Initiation & Centerline Identification ¹	1	\$3,500	\$3,500
2	Field Mapping Development & Routing Setup	500	\$12	\$6,005
3	RAC Mobilization & Setup	1	\$3,500	\$3,500
4	Onsite Field Pilot (days)	2	\$2,500	\$5,000
5	RAC Vehicle Pavement Distress Collection (test miles) ²	500	\$135	\$67,553
6	Pavement Evaluation, PCI Processing, & QA/QC (Modified ASTM D6433 using AI)	500	\$40	\$20,016
7	Pavement Width (lane miles)	762	\$12	\$9,150
8	Pavement Analysis, Budgetary Modeling, Multi-Year Plan Development (BOSS™) ³	1	\$15,000	\$15,000
9	Draft & Final Pavement Management Report	1	\$10,000	\$10,000
10	Project Management	1	\$9,781	\$9,781
Total Fee⁴				\$149,503

¹Assumes centerline file will be provided in GIS.

²Arterials and Collectors are two pass collection and residential are one pass collection.

³To be generated using the RAS **Budget Optimization Street Selector (BOSS™)** software. No software supply or maintenance fees are incurred.

⁴Bill monthly, lump sum based on percent complete for each task item.

	Optional Services			
11	Annual BOSS™ Pavement Analysis Update (After Year 1)	1	\$15,000	\$15,000
12	Cartegraph Data Load (pavement attribute and inspection data)	1	\$8,000	\$8,000
13	Cartegraph Scenario Builder Parameter Setup, Budgetary Modeling, & Configuration	1	\$14,500	\$14,500
14	Videologger & Data Viewer (Web Hosted)	1	\$9,000	\$9,000
15	Sign Inventory & Assessment (Lane Miles)	762	\$55	\$41,936
16	Sidewalk Inventory & Assessment (Lane Miles)	762	\$45	\$34,312
17	ADA Ramp Inventory & Assessment (Lane Miles)	762	\$40	\$30,499
18	Markings Inventory & Assessment (Lane Miles)	762	\$35	\$26,687
19	Striping Inventory & Assessment (Lane Miles)	762	\$40	\$30,499
20	Curb/Gutter Inventory & Assessment (Lane Miles)	762	\$45	\$34,312
21	Traffic Signal Inventory & Assessment (Lane Miles)	762	\$35	\$26,687
22	Senior Pavement Engineer Consulting Hours (Hourly)	1	\$275.00	\$275.00
23	Senior GIS Analyst Consulting Hours (Hourly)	1	\$150.00	\$150.00